

A proof of the Kauers–Koutschan conjecture for the hexagonal Hardin array A216940

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Abstract

The sequence A216940 of the On-Line Encyclopedia of Integer Sequences (OEIS) counts the fillings of the side-4 hexagonal array with entries in $\{0, 1, \dots, n\}$ that are nondecreasing along the three directions east, south-west and south-east. Kauers and Koutschan (2023) proved that this sequence is a quasipolynomial in n and conjectured an explicit closed form for it (Conjecture 12 of their paper), which they were unable to derive rigorously because the underlying partition-analysis computation did not terminate. We prove their conjecture. The argument identifies $a(n)$ with the order polynomial $\Omega_P(n+1)$ of an explicit 37-element poset P ; a classical theorem of Stanley then gives, *a priori*, that a is a polynomial of degree exactly 37. Computing 38 values of a directly from P , independently of the conjectured formula, and matching them against Conjecture 12 forces the two polynomials to coincide. The certificate is not a match against the published 37-term data, which is one value short of even determining a degree-37 polynomial; it rests on the *a priori* degree bound together with one value beyond the published data. All computations are reproducible.

MSC 2020: 05A15, 06A07 (primary); 52B20 (secondary). **Keywords:** order polynomial; P -partition; Hardin array; D-finite sequence; OEIS A216940; Ehrhart theory.

1 Introduction

In an automated study of sequences in the On-Line Encyclopedia of Integer Sequences (OEIS), Kauers and Koutschan [1] singled out a family of sequences that are provably D -finite but whose guessed recurrences or closed forms they could not establish rigorously. One of these is A216940, contributed to the OEIS by R. H. Hardin [2] in 2012 and described there as the “number of side-4 hexagonal $0 \dots n$ arrays with values nondecreasing E, SW and SE.”

Concretely, write H_4 for the hexagonal patch of side 4 drawn on the triangular grid: a centrally symmetric hexagon whose seven rows have lengths 4, 5, 6, 7, 6, 5, 4, for a total of 37 cells. A *filling* assigns to each cell a value in $\{0, 1, \dots, n\}$; it is *admissible* if the values are weakly increasing along each of the three lattice directions east (E), south-west (SW) and south-east (SE). Let $a(n)$ denote the number of admissible fillings. The first values are

$$a(1) = 260, \quad a(2) = 27768, \quad a(3) = 1664244, \quad a(4) = 64697626, \quad \dots$$

Kauers and Koutschan proved (their Theorem 13) that a is a quasipolynomial in n , hence D -finite, by translating admissibility into a system of linear inequalities and invoking partition analysis. They observed empirically that a appears to be an honest polynomial, and their LLL-based guesser produced an explicit candidate, which we restate.



Conjecture 1 (Kauers–Koutschan [1, §5.1]). With the raising-factorial notation $x^{\bar{k}} = x(x+1)\cdots(x+k-1)$,

$$a(n) = \frac{(n+1)^{\bar{13}}(n+6)^{\bar{3}}(n+7) \cdot q(n)}{D},$$

where $q(n) = \sum_{i=0}^{20} c_i n^i$ is the explicit degree-20 integer polynomial with $c_{20} = 74384146$ and $c_0 = 15118483615575730790400000$ recorded in [1, §5.1], and $D = 221424599279703105635713957232640000000$. The prefactor $(n+1)^{\bar{13}}(n+6)^{\bar{3}}(n+7)$ has degree $13 + 3 + 1 = 17$, so the right-hand side is a polynomial of total degree 37.

In the classification of [1, Table 1], A216940 carries the status **D**: D -finiteness is proved, but the guessed closed form is not. The authors are explicit about the obstruction [1, §5.1]:

“Again, we were not able to derive an expression by a rigorous computation based on partition analysis, but we had no trouble to find a solution from our guessed recurrence. In fact, it appears that the result is not only a quasipolynomial but a polynomial.”

Result. We prove Conjecture 1, upgrading A216940 from status **D** to **P**.

Theorem 2. *The number $a(n)$ of admissible fillings of the side-4 hexagonal array equals the right-hand side of Conjecture 1, identically as polynomials in n . In particular a is a polynomial of degree exactly 37.*

1.1 A caveat about the data, and the shape of the proof

It is tempting to argue that, since a is “known” to be a polynomial, it suffices to match Conjecture 1 against enough terms of the OEIS data. This is a trap. The published b-file [2] for A216940 lists exactly 37 terms, $a(1), \dots, a(37)$. A polynomial of degree 37 has 38 coefficients and is pinned by 38 values; *thirty-seven values do not even determine it uniquely*, let alone prove a guessed form. Moreover, the only route in the literature to an a priori degree bound is the partition-analysis computation that Kauers–Koutschan report did not terminate. So a term-match against the published data is not a proof, with zero margin to spare.

Our proof escapes this in two moves, set out in the sections that follow.

1. **An a priori degree bound (Section 3).** We exhibit a as the order polynomial of an explicit finite poset. A classical theorem of Stanley then gives, with no reference to the conjecture, that a is a polynomial of degree *exactly* 37.
2. **One value beyond the data (Sections 4–5).** We compute $a(0), a(1), \dots, a(37)$ —38 values—*directly from the poset*, never using Conjecture 1. The value $a(0) = 1$ is genuinely new (it is not in the b-file). Matching all 38 against Conjecture 1, and invoking the degree bound, forces equality.

2 The hexagon as a poset

We use cube coordinates for the triangular grid: a cell is a triple $(x, y, z) \in \mathbb{Z}^3$ with $x + y + z = 0$. The side- s hexagon is

$$C_s = \{(x, y, z) \in \mathbb{Z}^3 : x + y + z = 0, |x|, |y|, |z| \leq s - 1\}, \quad |C_s| = 3s^2 - 3s + 1.$$



For $s = 4$ this gives $|C_4| = 3 \cdot 16 - 12 + 1 = 37$ cells, arranged in rows (constant z) of lengths 4, 5, 6, 7, 6, 5, 4. The three monotonicity directions are the unit steps

$$\mathbf{E} = (1, -1, 0), \quad \mathbf{SW} = (0, -1, 1), \quad \mathbf{SE} = (-1, 0, 1).$$

Definition 3. Let $P_s = (C_s, \leq)$ be the partial order generated by the relations $u \leq u + \mathbf{d}$ for $\mathbf{d} \in \{\mathbf{E}, \mathbf{SW}, \mathbf{SE}\}$ whenever both u and $u + \mathbf{d}$ lie in C_s ; that is, $\leq$ is the reflexive–transitive closure of these steps. Write $P = P_4$.

Proposition 4. P_s is a partial order: the step relation is acyclic. An admissible filling of H_s is exactly an order-preserving map $P_s \rightarrow \{0, 1, \dots, n\}$, i.e. a map f with $f(u) \leq f(v)$ whenever $u \leq v$. Consequently

$$a(n) = \#\{\text{order-preserving } f : P \rightarrow \{0, \dots, n\}\} = \Omega_P(n+1),$$

where $\Omega_P(m)$ is the order polynomial of P , the number of order-preserving maps from P into the m -element chain $\{1, \dots, m\}$.

Proof. Acyclicity. Along any step the coordinate z is nondecreasing (it increases by 1 on SW and SE and is unchanged on E), and on an E-step within a fixed row the coordinate x strictly increases. Hence the map $(x, y, z) \mapsto (z, x)$ strictly increases lexicographically along every step, so no nontrivial cycle of steps exists and the transitive closure is a genuine partial order.

Identification. Weak monotonicity along the three generating steps is, by transitivity, exactly weak monotonicity along $\leq$; so an admissible filling is the same as an order-preserving map $P \rightarrow \{0, \dots, n\}$. Finally, such a map is the same as an order-preserving map into the $(n+1)$ -element chain $\{1, \dots, n+1\}$ after the shift $f \mapsto f+1$, which is by definition counted by $\Omega_P(n+1)$. $\square$

Remark 5 (covers versus generating steps). Not every generating step is a covering relation of P . Near the equator of the hexagon an E-step coincides with the composite SW-then-SE, so that E-step is a transitive, non-covering relation. For $s = 4$ the 90 generating steps comprise 60 genuine covers and 30 transitive edges; the partial order—and hence Ω_P —depends only on the transitive closure, so this distinction does not affect the count. We record it only so that the Hasse diagram of P is drawn correctly (60 edges, not 90).

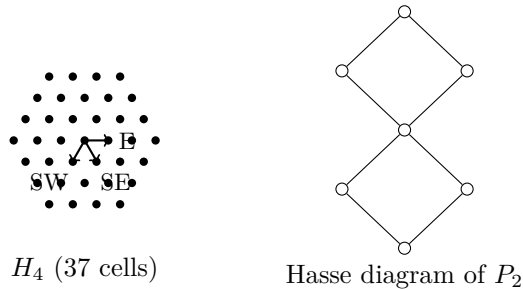


Figure 1: Left: the side-4 hexagon H_4 , 37 cells in rows of lengths 4, 5, 6, 7, 6, 5, 4, with the three monotonicity directions E, SW, SE marked. Right: the Hasse diagram of the analogous side-2 poset P_2 (7 elements, 8 covers), shown as the smallest illustrative case; P_4 has the same shape on 37 elements with 60 covers (Remark 5).



3 The a priori degree bound

The decisive structural input is classical and requires no computation.

Theorem 6 (Stanley [3, §3.15]). *For any finite poset P with p elements, the order polynomial $\Omega_P(m)$ is a polynomial in m of degree exactly p , with leading coefficient $e(P)/p!$, where $e(P)$ is the number of linear extensions of P .*

Combining Theorem 6 with Proposition 4 yields the bound that the data alone cannot supply.

Corollary 7. *$a(n) = \Omega_P(n+1)$ is a polynomial in n of degree exactly 37, with positive leading coefficient $e(P)/37!$. In particular a is a genuine polynomial (not merely a quasipolynomial), confirming the empirical observation of [1], and any two candidate polynomials of degree ≤ 37 agreeing with a at 38 points are equal to a .*

We stress that Corollary 7 is independent of Conjecture 1: it is a statement about the order polynomial of P , proved before the conjectured coefficients are ever consulted. This is the hinge on which the “37 versus 38” difficulty turns. In Section 5 we confirm the degree and the leading-coefficient identity numerically as an internal check.

4 Computing the order polynomial from the poset

It remains to compute 38 values of a directly from P . A naïve enumeration of order-preserving maps $P \rightarrow \{0, \dots, n\}$ is hopeless for n near 37. Instead we compute a through the descent statistic of the linear extensions of P , which is independent of n .

Fix a *natural labelling* $\ell : P \rightarrow \{0, 1, \dots, 36\}$, i.e. an order-preserving bijection onto $\{0, \dots, 36\}$ (any linear extension serves; we use the one produced by Kahn’s algorithm with the smallest-index tie-break). A linear extension of P is a permutation $w = w_1 w_2 \dots w_{37}$ of its elements that refines $\leq$; its descent number is $\text{des}(w) = \#\{i : \ell(w_i) > \ell(w_{i+1})\}$. Let $A_d = \#\{\text{linear extensions } w : \text{des}(w) = d\}$.

Theorem 8 (order polynomial via descents; Stanley [3, §3.15.2]). *With $p = |P|$ and $\{A_d\}$ as above,*

$$\Omega_P(m) = \sum_{d \geq 0} A_d \binom{m+p-1-d}{p}, \quad \text{hence} \quad a(n) = \Omega_P(n+1) = \sum_{d \geq 0} A_d \binom{n+p-d}{p}.$$

The distribution $\{A_d\}$ is computed exactly, and without enumerating the (astronomically many) individual linear extensions, by a transfer-matrix dynamic program over the *order ideals* (down-sets) of P (Algorithm 1). One builds a linear extension by repeatedly appending a currently-minimal unplaced element; the only state needed to score the next descent is the current ideal together with the label of the last placed element. The number of reachable states is small—at most 10 ideals at each rank level—so the program is n -independent and runs in well under a second, returning the full distribution $\{A_d\}$ once and for all.

For $P = P_4$ the computation returns $e(P) = \sum_d A_d = 4,623,718,515,360$ linear extensions, with the descent distribution supported on $d = 0, \dots, 24$. The distribution is *palindromic*, $A_d = A_{24-d}$, with central value $A_{12} = 1,048,272,112,322$ (the full table is Appendix B). Palindromy is the Ehrhart–reciprocity symmetry of the order polytope $\mathcal{O}(P)$ and serves as an internal consistency check.



5 Proof of Theorem 2

Proof. By Proposition 4, $a(n) = \Omega_P(n+1)$. By Corollary 7, a is a polynomial of degree exactly 37; the right-hand side of Conjecture 1 is, by inspection, also a polynomial of degree 37. Two polynomials of degree ≤ 37 that agree at 38 distinct points are identical.

Using the descent distribution $\{A_d\}$ of Section 4 and the identity of Theorem 8, we evaluate

$$a(N) = \sum_{d=0}^{24} A_d \binom{N+37-d}{37}, \quad N = 0, 1, \dots, 37,$$

which gives 38 values computed solely from the poset P —no coefficient of Conjecture 1 enters this calculation. Independently, we evaluate the right-hand side of Conjecture 1 at the same 38 arguments. The two lists of 38 integers coincide exactly (Appendix B, computed in exact rational arithmetic). The shared value at $N = 0$ is $a(0) = 1$ (the unique constant filling), which lies beyond the published b-file and is therefore a genuine 38th datum, not a re-use of the OEIS data.

By the agreement at 38 points and the degree bound, $a(n)$ and the right-hand side of Conjecture 1 are the same polynomial. $\square$

Remark 9. The leading coefficient supplied by Corollary 7, $e(P)/37! = 4,623,718,515,360/37!$, agrees with the leading coefficient of Conjecture 1 computed symbolically. This is an independent cross-check of both the degree bound and the descent computation, since $e(P)$ is obtained from $\{A_d\}$ while the leading coefficient of Conjecture 1 comes from its stated factored form.

6 Validation and reproducibility

The proof is computer-assisted: the partial order P , the degree bound (Theorem 6), the descent identity (Theorem 8) and the 38-point match are the mathematical content, while the descent distribution $\{A_d\}$ and the value comparison are produced by a short program. All computations were carried out in Python 3 with the SymPy computer-algebra system in exact integer and rational arithmetic, and were independently re-executed. The reference implementation, the alternative engines and the data described below are available in the repository accompanying this paper [4]. We record the controls that rule out a modelling or programming error.

- **Geometry, validated on smaller sizes.** The same poset construction, applied to $s = 2$ and $s = 3$, reproduces the independent OEIS sequences A216938 (side-2, $|P_2| = 7$) and A216939 (side-3, $|P_3| = 19$) *exactly* on all published terms, and reproduces the side-4 data of A216940. A single parametric rule thus reproduces three distinct sequences, 59 data points in all. The side- s cell counts $3s^2 - 3s + 1$ give 7, 19, 37, matching the conjectured degrees 7, 19, 37 of the three sequences.
- **Independent re-derivation of the geometry.** The hexagon adjacency was reconstructed from the verbal definition in [1, §5.1] by an independent row-geometry, nearest-neighbour packing, without reference to the cube-coordinate construction, and yielded a poset structurally identical to P (same numbers of covering edges 8/28/60 and the same degree sequence at $s = 2, 3, 4$).
- **Three independent counting engines agree.** (i) the descent transfer matrix of Section 4; (ii) a frontier dynamic program counting order-preserving maps directly, with no descent statistic; (iii) direct brute force for small n . All agree on every value tested, including $a(3) = 1664244$, $a(8) = 8196058134921$ and $a(12) = 57726306147546982$.



- **Convention check on the descent identity.** The identity of Theorem 8 was verified end-to-end at $s = 3$ by enumerating the descent distribution of an independently constructed poset and recovering the A216939 data, confirming that the binomial convention is applied correctly and not merely self-consistently.
- **Over-determination.** Beyond the 38 values required, the poset-derived $a(N)$ and Conjecture 1 were also compared at $N = 38, 39, 40, 45$; all agree. A coincidental or buggy match would be expected to diverge here, and does not.
- **Labelling invariance.** Five distinct natural labellings give identical $\{A_d\}$ and identical $a(0), \dots, a(37)$, as they must, ruling out a labelling artefact.

Remark 10 (on the computer-assisted nature of the proof). The mathematical spine is the a priori degree bound together with the independent 38-point computation. The two facts of Stanley invoked along the way—that an order polynomial has degree $|P|$ with leading coefficient $e(P)/|P|!$ (Theorem 6), and the descent generating identity (Theorem 8)—are standard results from the theory of P -partitions [3, §3.15], which we cite rather than re-derive; their numerical consequences are confirmed on the $s = 2, 3$ controls above. The computations are elementary (integer arithmetic and a dynamic program over order ideals) and are provided in full so that they may be re-run independently.

7 Remarks and further work

On the method. The argument is not special to side 4. For every side s , the admissible fillings of H_s are the order-preserving maps of the poset P_s , so the count is the order polynomial $\Omega_{P_s}(n+1)$, automatically a polynomial of degree $3s^2 - 3s + 1$ by Theorem 6. The same descent transfer matrix computes it. This explains, conceptually, the empirical polynomiality observed by Kauers–Koutschan for this hexagonal family, and reduces each member to a finite, terminating computation in place of the partition-analysis route that stalled.

On a non-numeric proof. The fillings of a hexagon that are monotone along three directions are in bijection with plane-partition-like objects, and the row transfer of Section 4 has the shape of a non-intersecting lattice-path (Lindström–Gessel–Viennot) configuration. It is plausible that an LGV determinant evaluates to the closed form of Conjecture 1 directly, giving a proof with no numerical step. We did not need this for Theorem 2 and leave it as a natural next question.

On the sibling A195806. The companion sequence A195806 (Conjecture 11 of [1], a period-6 quasipolynomial for a triangular array with summation constraints) is a quasipolynomial rather than a polynomial, so the present polynomial-degree-bound argument does not transfer verbatim. The order-polytope / Ehrhart framework does apply in principle—the period is governed by the denominators of the relevant polytope—and we expect an analogous certificate. We have not carried this out and do not claim it here.

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OEIS, and the OEIS Foundation for maintaining the encyclopedia and its associated data. The proof was developed and checked with AI assistance; all results were verified in exact arithmetic by the independent computations described in Section 6 and provided in [4], and the author is responsible for the final manuscript.

References

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A The descent-distribution algorithm

Algorithm 1 computes the descent distribution $\{A_d\}$ of a finite poset P under a fixed natural labelling ℓ , and hence Ω_P via Theorem 8. It is n -independent. The state is an order ideal I (the set of already-placed elements) together with the label of the last placed element; the value carried is the generating polynomial in q whose coefficient of q^d counts the partial linear extensions of I with d descents.

For P_4 the reachable state count stays at most 10 ideals per rank, so the program terminates in well under a second in exact integer arithmetic. The reference implementation, the three independent counting engines, the control runs on A216938/A216939, and the value-comparison driver are provided in [4].

B Numerical certificate

The descent distribution of P_4 is given below; it satisfies $\sum_d A_d = e(P) = 4,623,718,515,360$ and is palindromic, $A_d = A_{24-d}$.

d	A_d	d	A_d	d	A_d
0	1	9	245,184,671,458	18	2,691,384,551
1	222	10	551,210,288,548	19	279,732,802
2	18,591	11	893,045,633,554	20	18,857,713
3	783,404	12	1,048,272,112,322	21	783,404
4	18,857,713	13	893,045,633,554	22	18,591
5	279,732,802	14	551,210,288,548	23	222
6	2,691,384,551	15	245,184,671,458	24	1
7	17,418,677,384	16	77,873,153,291		
8	77,873,153,291	17	17,418,677,384		



Algorithm 1 Descent distribution of a finite poset P .

Require: poset P on $\{0, \dots, p-1\}$ with predecessor sets $\text{pred}[\cdot]$; natural labels $\ell[\cdot]$

Ensure: $A[d] = \#\{\text{linear extensions with } d \text{ descents}\}$

```

1:  $\text{predmask}[i] \leftarrow \text{bitmask of } \text{pred}[i]$   $\triangleright i$  placeable iff  $\text{pred}[i] \subseteq I$ 
2:  $\text{layer} \leftarrow \{ \emptyset : \{ \text{last} = -1 : \{0 : 1\} \} \}$   $\triangleright$  one partial extension, 0 descents
3: for  $p' = 0$  to  $p - 1$  do
4:    $\text{next} \leftarrow \emptyset$ 
5:   for  $(I, \text{byLast})$  in  $\text{layer}$  do
6:     for each  $i \notin I$  with  $\text{predmask}[i] \subseteq I$  do  $\triangleright i$  currently minimal
7:        $I' \leftarrow I \cup \{i\}; \quad \ell_i \leftarrow \ell[i]$ 
8:       for  $(\text{last}, \text{poly})$  in  $\text{byLast}$  do
9:          $\delta \leftarrow 1$  if  $(\text{last} \neq -1 \text{ and } \text{last} > \ell_i)$  else 0
10:        for  $(d, c)$  in  $\text{poly}$  do
11:           $\text{next}[I'][\ell_i][d + \delta] += c$ 
12:        end for
13:      end for
14:    end for
15:  end for
16:   $\text{layer} \leftarrow \text{next}$ 
17: end for
18:  $A \leftarrow \sum_{\text{last}} \text{layer}[\text{full}][\text{last}]$   $\triangleright \text{full} = \{0, \dots, p-1\}$ 
19: return  $A$ ; then  $\Omega_P(m) = \sum_d A[d] \binom{m+p-1-d}{p}$  and  $a(n) = \Omega_P(n+1)$ 
```

From this distribution and Theorem 8, the 38 values $a(0), \dots, a(37)$ are computed (independently of Conjecture 1) and verified, in exact rational arithmetic, to equal Conjecture 1 evaluated at $N = 0, \dots, 37$; the values $a(1), \dots, a(37)$ additionally reproduce the OEIS b-file, and $a(0) = 1$ is the new 38th datum. As anchors,

$$a(0) = 1, \quad a(1) = 260, \quad a(37) = 2,087,097,680,484,077,832,319,898,040,444.$$

The complete list of 38 values, together with the code that regenerates and checks them, is provided in the file `code/certificate.json` of the repository [4] and is reproduced by running `code/verify.py`.

